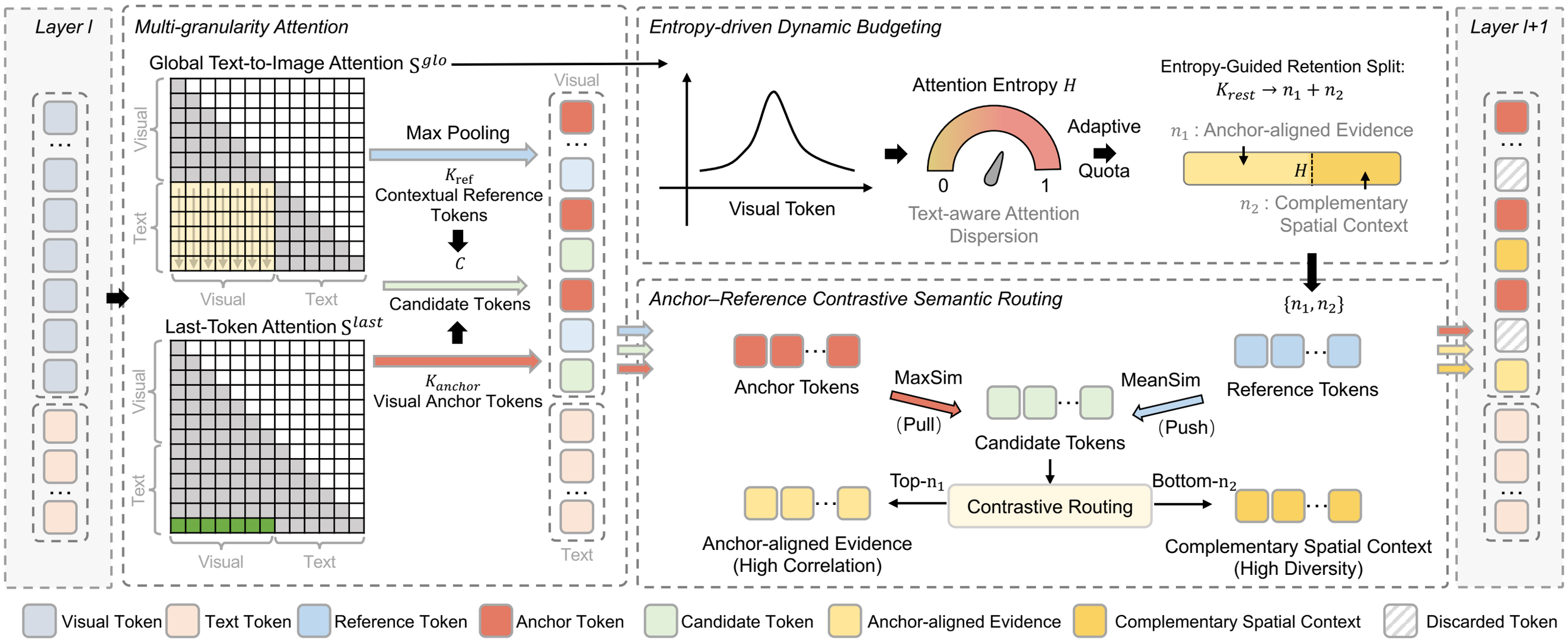

Layer l
Multi-granularity Attention
Global Text-to-Image Attention S^{glo}
Visual
Text
Max Pooling
K_{ref}
Contextual Reference Tokens
C
Candidate Tokens
Last-Token Attention S^{last}
K_{anchor}
Visual Anchor Tokens
Entropy-driven Dynamic Budgeting
Visual Token
Attention Entropy H
0
1
Text-aware Attention Dispersion
Adaptive
Quota
Entropy-Guided Retention Split:
$K_{rest} \rightarrow n_1 + n_2$
n_1 : Anchor-aligned Evidence
H
n_2 : Complementary Spatial Context
$\{n_1, n_2\}$
Anchor–Reference Contrastive Semantic Routing
Anchor Tokens
MaxSim
(Pull)
Candidate Tokens
MeanSim
(Push)
Reference Tokens
Top-n_1
Contrastive Routing
Bottom-n_2
Anchor-aligned Evidence
(High Correlation)
Complementary Spatial Context
(High Diversity)
Layer l+1
Visual Token
Text Token
Reference Token
Anchor Token
Candidate Token
Anchor-aligned Evidence
Complementary Spatial Context
Discarded Token